



Figure 1: Spacewalk architecture

- Do server inventory
- Work as a kickstart server for new server builds
- Manage virtual guests
- The minimum requirements for the Spacewalk server are:
 - X86 server with 2GB RAM
 - CentOS/Fedora/RHEL/SUSE/Debian/OL Linux
 - Either Oracle or PostgreSQL database
 - Storage for DB: approximately 1GB per channel which has 5000 packages
 - Storage for packages (for repos): approximately 6GB per channel
- An up-to-date OS - I tried the installation with Centos 6
- Ports 5222, 5269 and 443,80 should have a firewall
- Access to the Internet to install RPMs/packages from the Spacewalk repo
- Firewall ports used in Spacewalk include:
 - 69: TFTP (PXE provisioning)
 - 80: Spacewalk Web interface
 - 443: Spacewalk Web interface (SSL)
 - 4545: Spacewalk monitoring
 - 5222: If you plan to push actions to client systems
 - 5269: If you push actions to a Spacewalk proxy server
 - 9055: Oracle XE Web access

Installation

First, we need to install the RPM mentioned below to create repos which can point to Spacewalk RPMs. You can download the RPM from http://yum.spacewalkproject.org/2.3/RHEL/6/x86_64/ and install it with the following command:

```
#rpm -ivh spacewalk-repo-2.3-4.el6.noarch.rpm
```

Or you can use download and install it at one shot by using the following command:

```
#rpm -Uvh http://yum.spacewalkproject.org/2.3/RHEL/6/x86_64/spacewalk-repo-2.3-4.el6.noarch.rpm
```

This RPM will create two repos. One for Spacewalk binaries and the other for source packages.

Apart from these, you will need a few more OS RPMs to install. Hence, we will create one more repo for EPEL (Extra Packages for Enterprise Linux), as follows:

In EPEL 6 (for RHEL 6 and CentOS 6)

```
#rpm -Uvh https://dl.fedoraproject.org/pub/epel/epel-release-latest-6.noarch.rpm
```

In EPEL 7 (For RHEL 7 and CentOS 7)

```
#rpm -Uvhhttps://dl.fedoraproject.org/pub/epel/epel-release-latest-7.noarch.rpm
```

This will create two more repos – epel and epel-testing repos. As one of the pre-requisites, we need a database to store data. Here, I used Oracle-XE for testing.

We can install the following packages to create Oracle related repos:

```
#rpm -Uvh http://spacewalk.redhat.com/yum/1.4/RHEL/6/x86_64/spacewalk-client-repo-1.4-1.el5.noarch.rpm
#rpm -Uvhhttp://spacewalk.redhat.com/yum/1.4/RHEL/6/x86_64/spacewalk-repo-1.4-1.el5.noarch.rpm
```

Once done, please ensure all repos are accessible by using the following command:

```
#yum clean all
#yum repolist
```

Oracle installation

We can download Oracle and the following few required RPMs from <http://www.oracle.com/technology/software/products/database/xel/htdocs/102xe1nsoft.html>:

- oracle-xe-univ
- oracle-instantclient11.2-basic
- oracle-instantclient11.2-sqlplus

During the download, create the following user and group:

```
#groupadd dba
#useradd -g dba oracle
```

While installing Oracle, it may prompt you for *glibc* and *libaio* packages as Oracle needs 32-bit versions even on a 64-bit platform. So just download them and keep them aside to use if needed. Now, we install Oracle:

```
#rpm -Uvh oracle-xe-univ
#rpm -Uvh oracle-instantclient
#rpm -Uvh oracle-instantclient
#yum install oracle-lib-compat
```